

## Code-Layout, Readability And Reusability

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|----------------------|----------------------|
| <b>Date</b>          | 06 July 2026         |
| <b>Project Title</b> | <b>Rising Waters</b> |
| <b>Maximum Marks</b> | 5 Marks              |

### Code layout checklist:

| S.No | Code quality parameter    | description                                   | Followed (yes / No / partial ) | Remarks                                                                                 |
|------|---------------------------|-----------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | yes                            | Used consistent 4-space indentation following Python standards.                         |
| 2    | Proper File Structure     | Files and folders are logically organized     | yes                            | Organized into templates, static, models, dataset, and application files.               |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | yes                            | Used descriptive names such as rainfall, water_level, and prediction_result.            |
| 4    | Function/Method Names     | Functions are descriptively named             | yes                            | Functions like train_model(), predict_flood(), and preprocess_data() are clearly named. |
| 5    | Code Comments             | Inline and block comments explain log c       | yes                            | Added comments to explain important sections and functions.                             |
| 6    | Modular Design            | Code is split into reusable functions/modules | yes                            | Data preprocessing, model training, prediction, and Flask routes are separated.         |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | yes                            | Removed unnecessary and duplicate code to improve maintainability.                      |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | partial                        | Basic exception handling and input validation are                                       |

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|  |  |  |  | implemented; can be enhanced further. |
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### Reusable Components / Modules :

| S.No | Component / Module Name   | Language / Technology | Where Reused                            | Reusability level |
|------|---------------------------|-----------------------|-----------------------------------------|-------------------|
| 1    | Data Preprocessing Module | Python, Pandas        | Data cleaning and feature preparation   | High              |
| 2    | Flood Prediction Model    | Python, Scikit-learn  | Prediction for different user inputs    | High              |
| 3    | Flask Routes              | Python, Flask         | Handles multiple web requests           | High              |
| 4    | HTML Templates            | HTML, CSS, Bootstrap  | Home page, prediction page, result page | Medium            |
| 5    | Model Loading Module      | Python, Joblib        | Loads the trained model for predictions | High              |

### Overall Code Quality Assessment :

| Aspect                    | Rating (1-5) | Comments                                                                                                          |
|---------------------------|--------------|-------------------------------------------------------------------------------------------------------------------|
| Code layout and structure | 5            | Well-organized project with separate folders for datasets, models, templates, static files, and application code. |
| Readability               | 5            | Used meaningful variable names, proper indentation, and consistent coding style.                                  |
| Reusability               | 5            | Modular functions for preprocessing, model prediction, and Flask routes can be reused easily.                     |
| Documentation / comments  | 4            | Important functions are documented with comments; additional documentation can be added if needed.                |
| Overall score             | 4.8          | The project follows good coding practices and is maintainable, readable, and reusable.                            |